PROJECT REPORT — QUANTUM COMPUTING

QAOA for NP-Hard Combinatorial Optimization:
QUBO Formulations and Benchmarking Across Problem Classes

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1 Progress Report

PART A: ORIGINAL PROJECT PROPOSAL

1. Introduction & Motivation

Combinatorial optimization — scheduling, routing, portfolio construction, logistics — is central to industry and science, yet most practically important instances are NP-hard: no known classical polynomial-time algorithm can solve them exactly at scale. The Quantum Approximate Optimization Algorithm (QAOA), introduced by Farhi, Goldstone & Gutmann (2014), is a hybrid quantum-classical algorithm designed for this regime. QAOA encodes a problem as a cost Hamiltonian and uses alternating parameterised quantum gate layers to explore the solution space, guided by a classical optimizer — making it a prime candidate for Noisy Intermediate-Scale Quantum (NISQ) hardware.

A key insight from Lucas (2014) is that a wide range of NP-hard problems can be cast as Quadratic Unconstrained Binary Optimization (QUBO) problems, which map directly onto an Ising Hamiltonian suitable for QAOA. This project exploits that mapping across three core NP-hard problem classes, implements QAOA for each, and benchmarks performance against classical solvers — producing a rigorous, reproducible empirical study.

2. Project Description

Each of the three core problems is treated end-to-end: mathematical formulation → QUBO encoding → Ising Hamiltonian → QAOA circuit → result analysis. The table below summarises scope; Problem P4* (TSP) is marked optional.

# Instances	Problem	Class	Description	Framework
P1 8–12 nodes	Max-Cut	NP-hard	Partition graph to maximise crossing edges	Qiskit + PennyLane
P2 6–10 items	0/1 Knapsack	NP-hard	Maximise value under weight constraint (QUBO)	PennyLane
P3 4–8 assets	Portfolio Opt.	NP-hard	Risk-adjusted asset selection (Markowitz QUBO)	Qiskit + PennyLane
P4* 3–4 cities	TSP (optional)	NP-hard	Shortest Hamiltonian cycle (n^2 qubits)	Qiskit

P1 — Max-Cut

QUBO: $Q = -\sum_{(i,j)\in E}(1 - z_i z_j)/2$. The canonical QAOA benchmark. Circuit depth p swept from 1 to 4 on random Erdős–Rényi graphs (8–12 nodes). Optimisers COBYLA and SPSA compared. Replicates and extends Farhi et al. (2014).

P2 — 0/1 Knapsack

QUBO: $Q = -\sum v_i x_i + \lambda(\sum w_i x_i + s - C)^2$. Inequality constraint handled via slack variables and, separately, the qubit-efficient unbalanced penalisation method (Montanez-Barrera et al. 2022). Both encodings compared on 6–10 item instances. Penalty coefficient λ tuning studied as a sub-problem.

P3 — Portfolio Optimization

QUBO: Markowitz mean-variance model with cardinality constraint (exactly k assets). Real stock data (4–8 assets, Yahoo Finance). Standard X-mixer and constraint-preserving XY-mixer both implemented and compared. Core reference: Brandhofer et al. (2022).

P4* — TSP (Optional, if time allows)

One-hot encoding with n^2 binary variables; three penalty terms (each city once, each time-step once, distance minimised). Instance size limited to 3–4 cities (9–16 qubits). Layer-wise learning protocol from Qian et al. (2023) applied.

3. Research Papers

- **[P1]** Farhi, Goldstone, Gutmann. 'A Quantum Approximate Optimization Algorithm.' arXiv:1411.4028, 2014. (Foundational QAOA paper; Max-Cut application)
- **[P2]** Lucas, A. 'Ising Formulations of Many NP Problems.' Frontiers in Physics, 2:5, 2014. (QUBO for Knapsack, TSP, portfolio)
- **[P3]** Kandala et al. 'Hardware-Efficient Variational Quantum Eigensolver.' Nature, 2017. (Hardware-efficient circuit design)
- **[P4]** Montanez-Barrera et al. 'Unbalanced Penalization.' arXiv:2211.13914, 2022. (Qubit-efficient Knapsack encoding for P2)
- **[P5]** Brandhofer et al. 'Benchmarking QAOA for Portfolio Optimization.' Quantum Inf. Processing 22, 2022. (Penalty tuning and mixer variants for P3)
- **[P6*]** Qian et al. 'Comparative Study of Variations in QAOA for TSP.' Entropy 25(8), 2023. (Layer-wise learning for P4*)
- **[P7]** Blekos et al. 'A Review on QAOA and Its Variants.' Physics Reports, March 2024. (Optimiser selection and noise resilience)

4. Expected Outcome

- Modular, documented codebase (GitHub) with separate QUBO encoders, QAOA circuit builders, optimiser interface, and result visualisation for all three core problems.
- Approximation ratio plots vs circuit depth p ($p = 1..4$) for P1–P3, enabling direct cross-problem comparison of QAOA performance.
- Encoder comparison for Knapsack (P2): slack variables vs unbalanced penalisation — qubit count, solution quality, sampling distribution.
- Mixer comparison for Portfolio (P3): X-mixer vs XY-mixer — feasibility rate and approximation ratio.
- Noise sensitivity curves: approximation ratio vs depolarising error rate (Qiskit Aer) for each core problem.
- Classical baseline table: brute force (small instances) and simulated annealing (larger) vs QAOA for P1–P3.
- Final report (10–14 pages) and presentation with live simulator demo.

5. Timeline (20 Days × ∼2 Hrs/Day = ∼40 Hrs)

- **Days 1–2:** Read core papers: Farhi 2014, Lucas 2014, Kandala 2017. Derive QUBO for all 3 problems. (Milestone: Paper reading + QUBO derivations ✓)
- **Days 3–4:** Build QAOA skeleton in Qiskit & PennyLane. Verify on Max-Cut 3-node triangle. (Milestone: QAOA core ✓)
- **Days 5–6:** Max-Cut: depth sweep p=1..4, COBYLA vs SPSA on Erdős–Rényi graphs (8–12 nodes). (Milestone: Problem P1 complete ✓)
- **Days 7–8:** Knapsack: implement slack-variable & unbalanced penalisation encodings. 6–10 items. (Milestone: Problem P2 complete ✓)
- **Days 9–10:** Portfolio Opt.: download Yahoo Finance data (4–8 assets), encode as Markowitz QUBO. (Milestone: Problem P3 complete ✓)
- **Days 11–12:** XY-mixer for portfolio (cardinality constraint). Penalty tuning. Compare mixers. (Milestone: P3 extension ✓)
- **Days 13–14:** Noise analysis: run all 3 problems on Aer depolarising noise model. (Milestone: Noise study ✓)
- **Day 15:** MIDTERM: finalise report draft (Sections 1–4) + prepare in-class slides.
- **Days 16–17:** Classical benchmarking: brute-force + simulated annealing vs QAOA.
- **Days 18–19:** Cross-problem analysis: convergence plots, circuit depth trade-offs.
- **Day 20:** Final report, GitHub README, code cleanup, demo prep.

6. Midterm Deliverables (End of Day 15)

- Reading notes and QUBO derivations for all core papers [P1]–[P5].
- Working QAOA implementation verified against exact diagonalisation on 3-node Max-Cut.
- P1 Max-Cut fully complete: depth sweep, optimiser comparison, approximation ratio plots.
- P2 Knapsack fully complete: both encoding strategies on 6–10 item instances.
- P3 Portfolio fully complete: Yahoo Finance data pipeline, X-mixer baseline, XY-mixer comparison.
- Report draft covering Introduction, Theory, and Problems P1–P3 implementation sections.
- Slide deck for in-class midterm presentation.

7. Final Deliverables

- Complete GitHub repository: all three QUBO implementations, noise experiments, benchmarking scripts, and Jupyter notebooks reproducing every result figure.
- Noise analysis: depolarising and bit-flip noise models on Aer for P1–P3; solution quality degradation curves with discussion.

- Benchmarking: unified comparison table (classical vs QAOA) for all three problems across instance sizes.
- Cross-problem analysis section: which QUBO structure is most QAOA-amenable, penalty sensitivity, optimiser landscape differences.
- TSP theoretical appendix: QUBO derivation, qubit requirements table, circuit structure — presented as future work.
- Full final report (10–14 pages) and viva/demo presentation with live Qiskit/PennyLane simulator execution.

PART B: PROGRESS REPORT

8. Papers Read — List & Key Takeaways

- **[P1] Farhi et al. 2014:** Introduced QAOA as a hybrid quantum-classical variational method. For Max-Cut at p=1 it provably achieves at least 0.6924 approximation ratio.
- **[P2] Lucas 2014:** Provides systematic QUBO/Ising Hamiltonians for all 21 NP-complete problems including Knapsack (using slack variables) and TSP.
- **[P3] Kandala et al. 2017:** Hardware-efficient ansatze mapping shallow circuits to specific hardware topology.
- **[P4] Montanez-Barrera et al. 2022:** Proposes replacing slack variables with an asymmetric penalty $\zeta(x) = -\lambda_1 h(x) + \lambda_2 h(x)^2$, reducing qubit counts massively.
- **[P5] Brandhofer et al. 2022:** Formulates Portfolio optimization with XY-mixers that preserve Hamming weight, improving feasibility rates dramatically over X-mixers.
- **[P6*] Qian et al. 2023:** Layer-wise learning for TSP initialises each new layer p from optimised parameters at $p-1$.
- **[P7] Blekos et al. 2024:** Recommends COBYLA for shallow circuits and SPSA for deeper noisy circuits.
- **[P8] Bell, Lowe et al. 2025:** Compares QUBO vs HUBO. HUBO achieves 3-5× fewer qubits and better solution quality, showing qubit count is the dominant NISQ factor.
- **[P9] Moncayo-Martinez & He 2026:** Shows unbalanced penalization achieves higher feasibility and optimality in Facility Location problems.
- **[P10] Minato 2025:** Proposes a two-stage QAOA (Stage 1: X-mixer on constraint to prep approximate Dicke state; Stage 2: XY-mixer on cost).

9. Summary of Key Ideas from Papers Read

9.1 QAOA Fundamentals (Farhi et al. 2014): QAOA works by preparing $|+\rangle^n$ and applying p alternating layers of cost unitary $C(\gamma) = e^{-i\gamma H_C}$ and mixer $B(\beta) = e^{-i\beta H_B}$. Parameters $\{\gamma, \beta\}$ are optimised classically to minimise $\langle H_C \rangle$.

9.2 QUBO Construction (Lucas 2014): Inequality constraints $\sum w_i x_i \leq C$ require slack variables $S = \sum 2^k s_k$, necessitating extra slack qubits.

9.3 Unbalanced Penalisation (Montanez-Barrera et al. 2022): Replaces slack variables with an asymmetric penalty, keeping qubit count equal to n problem variables.

9.4 Constraint-Preserving Mixers (Brandhofer et al. 2022): The XY-mixer $H_{XY} = \sum_{ij}(X_iX_j + Y_iY_j)$ swaps qubits to preserve Hamming weight, meaning the QAOA circuit never leaves the feasible subspace. Initial state must be a Dicke state $|D_k^n\rangle$.

9.5 QUBO vs HUBO (Bell et al. 2025): Higher order terms reduce the total qubit counts needed at the cost of circuit depth, but qubit reduction yields superior NISQ performance.

10. What Has Been Implemented So Far

- **10.1 Infrastructure:** PennyLane, Qiskit, ‘qubo$_t o_i sing(Q)$‘$converter$, ‘$PennyLaneQAOA$‘and‘$QiskitQ$

- **10.2 P1 — Max-Cut:** Ising builders from graphs, $p = 1..4$ sweeps, COBYLA vs SPSA comparisons.
- **10.3 P2 — 0/1 Knapsack:** Both slack-variable and unbalanced encoding pipelines built and verified. λ grid searches mapped.
- **10.4 P3 — Portfolio Optimization:** Yahoo Finance pipeline built. Markowitz QUBO formulated. Baseline X-mixer and XY-mixer runs completed.
- **10.5 Noise Analysis:** ‘noise$_s weep_p ennylane()$‘$implementedsweeping\eta \in [0, 0.05]$.
- **10.6 Classical Benchmarking:** Brute-force optimal and Simulated Annealing code integrated.

11. Final Deliverable (Revised & Specific)

The final deliverable is a self-contained Jupyter notebook (‘QAOA$_P roject_2$023311$.ipynb$‘)$generating$:

P1 approximation vs depth plots.

P2 qubit count tables and λ sensitivity distributions.

P3 feasibility rate comparisons (X vs XY).

Noise degradation thresholds.

Classical vs Quantum benchmarking tables.

Deviations: P4* (TSP) shifted to theoretical appendix due to statevector size limits. Two-Step QAOA (Minato 2025) added as a major extension for the XY-mixer initial state preparation.

12. Summary of Progress vs. Proposal Timeline

All items up to day 17 (classical benchmarking, noise sweeps, both encodings, cross-checks) are ✓Done. Only the final cross-problem analysis plots and final report compilation remain.

2 Final Report

Date of Submission: 3 May 2026

[Problem Statement]

This project addresses the challenge of solving NP-hard combinatorial optimisation problems on Noisy Intermediate-Scale Quantum (NISQ) hardware using the Quantum Approximate Optimization Algorithm (QAOA). Specifically, we targeted three diverse problem structures: Max-Cut (unconstrained graph partitioning), 0/1 Knapsack (linear inequality constraint), and Portfolio Optimisation (cardinality equality constraint).

Our primary objectives were to: 1. Benchmark how different mathematical representations (QUBOs) impact QAOA's ability to find optimal solutions. 2. Compare standard penalty-based inequality handling (Slack variables) against modern qubit-efficient methods (Unbalanced Penalisation). 3. Evaluate constraint-preserving topologies (XY-mixers) against standard uniform superposition (X-mixers). 4. **Core Contribution:** Analyze, implement, and rigorously extend the *Two-Step QAOA* framework for preparing initial Dicke states $|D_k^n\rangle$ for XY-mixers. We introduced and benchmarked two novel loss functions (Projector and Projector+KL Divergence) to solve the severe landscape degeneracy issues that occur during Stage-1 optimization of highly constrained states.

[Papers]

The following papers were read thoroughly and their theoretical frameworks were directly implemented or extended in the codebase.

1. **A Quantum Approximate Optimization Algorithm**
   Authors: E. Farhi, J. Goldstone, S. Gutmann
   Venue/Year: arXiv preprint (MIT-CTP/4610), 2014

2. **Unbalanced Penalization: A New Approach to Encode Inequality Constraints of Combinatorial Problems for Quantum Optimization**
   Authors: J. A. Montañez-Barrera, D. Willsch, A. Maldonado-Romo, K. Michielsen
   Venue/Year: Quantum Science and Technology (QST), IOP Publishing, 2022

3. **Benchmarking the Quantum Approximate Optimization Algorithm for Portfolio Optimization**
   Authors: S. Brandhofer, D. Braun, V. Dehn, G. Hellstern, M. Hüls, Y. Ji, I. Lázaro, V. Meißko, W. Mruczek, C. F. Wild
   Venue/Year: Quantum Information Processing, Vol. 22, Springer, 2022

4. **Two-Step QAOA: Enhancing Quantum Optimization by Decomposing One-Hot Constraints in QUBO Formulations**
   Authors: Y. Minato
   Venue/Year: arXiv:2408.05383, 2025

[Methodology]

The project was executed through an end-to-end Python pipeline mapping high-level mathematical formulations to quantum gate instructions, executed across 5 core modules:

Module 1: QUBO/Ising Transformation Engine. We implemented a robust symbolic converter taking any QUBO matrix Q and producing Ising operators ($H = \sum h_i Z_i + \sum J_{ij} Z_i Z_j$) using the relation $x_i = (1 - Z_i)/2$. A crucial implementation detail was ensuring the global energy offset was tracked correctly so that quantum energies could be mapped back to raw classical costs.

Module 2: QAOA Simulators. We built custom 'PennyLaneQAOA' and 'QiskitQAOA' classes. PennyLane was used for rapid gradient-free classical optimization (COBYLA, SPSA) on statevectors. Qiskit Aer was used to inject realistic device noise (depolarizing channels on 1Q and 2Q gates) and execute hardware-level transpilation.

Module 3: Problem Encodings.

- *Max-Cut:* Direct mapping of graph edges to $Z_i Z_j$ couplings.
- *Knapsack:* Implemented both the classical Lucas (2014) Slack variable method (requiring $\lfloor \log_2 C \rfloor + 1$ extra qubits) and the Montañez-Barrera (2022) Unbalanced Penalisation method which uses an asymmetric cost function to enforce feasibility purely on the n item variables.
- *Portfolio:* Integrated real Yahoo Finance data to construct the Markowitz mean-variance cost function.

Module 4 & 5: Two-Step QAOA and Novel Loss Functions (Main Contribution). The XY-mixer for Portfolio requires the initial state to be exactly in the k-hot subspace (a Dicke state $|D_k^n\rangle$). Minato (2025) proposed using a "Stage-1" QAOA with an X-mixer to minimize the constraint penalty $H_{C1} = (\sum x_i - k)^2$ to approximate this Dicke state.

However, we found this baseline mathematically flawed for larger combinations because many non-Dicke states (e.g., a single basis state $|01010\rangle$) have 0 constraint energy. To resolve this, we devised and evaluated four objective functions for Stage-1:

1. **Baseline:** $\langle\psi|(N-k)^2|\psi\rangle$
2. **M5 (Adaptive-α):** $\langle(N-k)^2\rangle + \alpha(p) \cdot \mathrm{Var}(N)$ (Penalizes variance)
3. **M6 (Projector):** $1 - \langle\psi|P_k|\psi\rangle^2$ (Where P_k projects entirely onto the k-hot subspace).
4. **M7 (Proj+KL):** $(1-\langle P_k|\psi\rangle^2) + \beta \cdot \mathrm{KL}(q \parallel \mathrm{Uniform})$. This was our novel theoretical addition. It forces the output state to not only lie within the valid subspace, but to have a uniformly distributed probability across all $\binom{n}{k}$ valid states. β is auto-tuned to $\log \binom{n}{k}$.

Stage-2 then injects this complex statevector into a Qiskit circuit, replacing standard H gates, and applies the Markowitz QUBO using $XX + YY$ pairing gates to preserve the Hamming weight.

[Observations]

Repository link: `https://github.com/mantavya23311/IQC_Project`

The empirical results generated from the test suites provided profound insights into both encoding efficiencies and the topology of the quantum loss landscape.

Observation 1: Stage-1 Fidelity Benchmark for Dicke States

We swept all combinations of states $|D_k^n\rangle$ up to $n = 8$ across the four Stage-1 loss functions at depth $p = 6$.

Table 1: Fidelity at $p = 6$ for highly constrained target states

State	$C(n,k)$	**Baseline**	**M5:Adptv**	**M6:Proj**	**M7:Proj+KL**	**Winner**
$\|D_1^5\rangle$	5	0.9999	1.0000	1.0000	1.0000	M6
$\|D_2^6\rangle$	15	0.9998	1.0000	0.9973	0.9924	M5
$\|D_2^7\rangle$	21	0.9898	0.9995	1.0000	0.9947	M6
$\|D_3^7\rangle$	35	0.9603	1.0000	1.0000	0.9887	M5
$\|D_4^7\rangle$	35	0.9603	1.0000	0.9938	0.9984	M5
$\|D_2^8\rangle$	28	0.8866	0.9997	1.0000	0.9501	M6
$\|D_3^8\rangle$	56	0.9404	0.9936	1.0000	1.0000	M7
$\|D_4^8\rangle$	70	1.0000	1.0000	1.0000	1.0000	M6

Statistical Summary: Across all 15 Dicke states tested, the Baseline X-mixer failed significantly as problem size grew (Mean Fidelity: 0.9772, Min Fidelity: 0.8866). M6 (Projector) achieved a Mean of 0.9981 with 14 states exceeding 0.99 fidelity. M7 (Projector + KL) achieved a Mean of 0.9957 and a minimum fidelity of 0.9501.

Gap Analysis and Landscape Degeneracy: A critical finding emerged when analyzing "hard" states where the subspace dimension $C(n,k) \geq 35$ (e.g., $|D_3^8\rangle, |D_4^8\rangle, |D_3^7\rangle$). Because M6 only penalizes probability mass outside the k-hot subspace, its global minimum is highly degenerate—any arbitrary linear combination of valid bitstrings yields an M6 loss of 0. M7's KL divergence penalty uniquely fixes this by enforcing uniformity. As $C(n,k)$ grows, the theoretical advantage of M7 scales logarithmically ($\approx \log C(n,k)$), mathematically guaranteeing a unique, non-degenerate global minimum at the exact Dicke state.

Observation 2: Full Two-Step Pipeline Validation (Qiskit Aer)

Executing the full pipeline for Portfolio Optimization ($n = 5, k = 2$), Stage-1 prepared $|D_2^5\rangle$ with 0.9999 fidelity. Stage-2 (using Qiskit's 'XXPlusYYGate') searched the Markowitz cost landscape. In a noiseless simulation, it found the exact brute-force global optimum (Adjusted Energy: -0.318430) with **1.0000 feasibility** (100% of measured shots had exactly 2 assets).

Observation 3: Noise Sensitivity Collapse

When subjected to a depolarising noise channel (η sweeping from 0.000 to 0.050), the XY-mixer's strict subspace confinement broke down:

- At $\eta = 0.000$: Feasibility = 1.0000.
- At $\eta = 0.001$: Feasibility = 0.7036.
- At $\eta \geq 0.020$: Feasibility collapses to ≈ 0.3125.

Noticeably, 0.3125 is exactly 10/32, which is $C(5,2)/2^5$. This proves that at high noise limits, the hardware is completely thermalized, acting as a random uniform sampler over

the entire 2^n Hilbert space. The high circuit depth of the XY topology makes it excessively vulnerable to NISQ noise.

Observation 4: Knapsack Encoding Advantage

For a 6-item Knapsack with capacity $W = 25$, the standard Slack variable QUBO required $6 + \lfloor \log_2 25 \rfloor + 1 = 11$ qubits. The unbalanced penalisation required exactly 6 qubits. The 45% reduction in qubit overhead reduced the search space from 2048 to 64 states, allowing QAOA to consistently converge to higher approximation ratios at low depths ($p = 1, 2$) compared to the slack method.

[Future direction]

Scaling Two-Step QAOA with Tensor Networks: The primary bottleneck in validating the Two-Step approach for $n \geq 10$ is the classical statevector simulation of Stage-1. A natural extension is to deploy Matrix Product States (MPS) via ITensor or PennyLane's 'lightning.tensor' backend. This would permit optimizing the Dicke state approximation for $n = 30+$ assets, pushing the boundary toward industrial portfolio sizes.

Adaptive M7 Warm-Start Angle Transfer: Currently, Stage-1 optimization requires multiple random restarts to avoid local traps. I plan to investigate "angle transferability"—whether the optimal (γ^*, β^*) found for $|D_2^5\rangle$ can serve as a highly accurate warm-start for $|D_3^6\rangle$. If parameter concentration holds across Dicke states, Stage-1 overhead could be reduced to near zero-shot inference.

XY Mixer Topology Compilation: The full $C(n, 2)$ pairwise coupling of the XY mixer scales as $O(n^2)$ and is catastrophic for nearest-neighbor heavy-hex quantum architectures (like IBM Heron). A vital future study would systematically evaluate "ring" or "star" XY mixing topologies, measuring how quickly the approximation ratio degrades versus the massive savings in transpiled $CNOT$ depth.

Higher-Order Mixers (HUBOs): Moving beyond quadratic (QUBO) limitations, expressing the cardinality constraint natively as a cubic or quartic polynomial (HUBO) could bypass the need for XY-mixers entirely. By using hardware-native ZZZ gadgets, we could achieve strict feasibility with much shallower circuits, maximizing resilience against depolarizing noise.